

Mixed-Effects Models for Analyzing Intensive Longitudinal Data on Suicidal Ideation in PTSD

Jiaxin Chen

The Ohio State University

08-06-2025

Outline

- 1 Background
- 2 Methods
- 3 Results
- 4 Conclusion
- 5 References

Suicidal Crises

- Suicide is the 11th leading cause of death in the United States, and has increased by over 35% since 1999.
- Preventing suicide is challenging because suicidal crises can escalate rapidly, and can occur when a clinician is not immediately available.
- Safety planning-type interventions help patients manage suicide risk independently in their daily lives.

Crisis Response Planning (CRP) vs. Self-Guided Safety Planning (SP)

CRP

- Therapist-guided and collaborative
- Begins with narrative assessment of a crisis
- Plan includes: warning signs, coping, reasons for living, support, ...
- Emphasizes personalization

SP

- Self-administered with minimal therapist input
- No narrative assessment
- Plan includes: warning signs, coping, distraction, help contacts, ...
- Completed on templated paper or digital forms

Purpose

- This study examined whether associations between frequency of plan use and key clinical outcome (suicide ideation) varied across these different plan types.

Participants and Procedures

- ① Participants included in the present analyses were 116 U.S. military personnel and veterans meeting diagnostic criteria for full or subthreshold PTSD.
- ② Participants were randomly assigned to either CRP or self-guided SP.
- ③ During treatment, participants received 4 ecological momentary assessment (EMA) alerts per day for 14 consecutive calendar days.

Measures: Suicidal Ideation

- 1 Momentary suicidal ideation (SI) was assessed at each EMA prompt using the first five items of the Scale for Suicidal Ideation (Beck et al., 1988).
- 2 It covered aspects like the wish to live, wish to die, balance of reasons for living versus reasons for dying, active suicidal ideation, and passive suicidal ideation.
- 3 The five items were summed to provide an overall score, with higher totals indicating more severe suicidal ideation.

Measures: Frequency of Plan Use

- ① At the end of each day, participants were asked to report how many times they had used their assigned plan (either CRP or self-guided SP) during the day.
- ② “How many times did you use your crisis response plan or self-guided safety plan today?”
- ③ Response options ranged from 0 to 10.

Between- and Within-person Plan Use Frequency

- To separate the between- and within-subject effect of use of plan, we created two additional variables.
- We first calculated the grand mean, by averaging over time points within subjects firstly, and then averaging across subjects. To be specific for a longitudinal study, we observe subjects $i = 1, 2, \dots, n$, and for each subject i , we make $j = 1, \dots, t_i$ observations. Then grand mean μ was calculated as $\mu = \frac{1}{n} \sum_{i=1}^n [\frac{1}{t_i} \sum_{j=1}^{t_i} x_{ij}]$
- we additionally split it into two (orthogonal) components: a between-subject means component, $\mu_i - \mu = \frac{1}{t_i} \sum_{j=1}^{t_i} x_{ij} - \mu$, and a within-subject deviations from those means component, $x_{ij} - \mu_i = x_{ij} - \frac{1}{t_i} \sum_{j=1}^{t_i} x_{ij}$.

Data Analysis: Suicidal Ideation

- 1 We applied a logistic mixed-effects model to explore the association between plan use frequency and the likelihood of SI (1= SI; 0 = no SI).
- 2 Fixed effects included time, within-person and between-person plan use frequency, treatment group (CRP or SP), and their interactions to assess if treatment condition moderated the association between plan use and SI.

Data Analysis: Suicidal Ideation

- 1 A random intercept was included to account for baseline differences in SI across individuals.
- 2 We also considered additional models controlling for potential confounders (alcohol consuming, places, social context, ...) provided at each EMA prompt.
- 3 None of our results changed with the inclusion of these covariates; thus, we reported the unadjusted model for ease of interpretation in this study.

Results: SI Model

Table 1: Logistic Mixed-Effect Model for Suicidal Ideation (SI)

Fixed effects	Estimate(SE)
Intercept	-1.051 (0.001)***
Time	-0.887 (0.001)***
Plan use freq. (within-subject)	0.449 (0.001)***
Plan use freq. (between-subject)	1.021 (0.001)***
Group (CRP)	-0.446 (0.001)***
Plan use freq. (within-subject) $\times$ Group (CRP)	-0.373 (0.001)***
Plan use freq. (between-subject) $\times$ Group (CRP)	-0.917 (0.001)***
Random effects	Variance
Intercept	28.38

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Participants were less likely to endorse SI over time during our experiment.

Table 1: Logistic Mixed-Effect Model for Suicidal Ideation (SI)

Fixed effects	Estimate(SE)
Intercept	-1.051 (0.001)***
Time	-0.887 (0.001)***
Plan use freq. (within-subject)	0.449 (0.001)***
Plan use freq. (between-subject)	1.021 (0.001)***
Group (CRP)	-0.446 (0.001)***
Plan use freq. (within-subject) $\times$ Group (CRP)	-0.373 (0.001)***
Plan use freq. (between-subject) $\times$ Group (CRP)	-0.917 (0.001)***
Random effects	Variance
Intercept	28.38

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Both within-person plan use frequency and between-person plan use frequency were significantly associated with SI.

Table 1: Logistic Mixed-Effect Model for Suicidal Ideation (SI)

Fixed effects	Estimate(SE)
Intercept	-1.051 (0.001)***
Time	-0.887 (0.001)***
Plan use freq. (within-subject)	0.449 (0.001)***
Plan use freq. (between-subject)	1.021 (0.001)***
Group (CRP)	-0.446 (0.001)***
Plan use freq. (within-subject) $\times$ Group (CRP)	-0.373 (0.001)***
Plan use freq. (between-subject) $\times$ Group (CRP)	-0.917 (0.001)***
Random effects	Variance
Intercept	28.38

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Both within-person plan use frequency and between-person plan use frequency were significantly associated with SI.
- Specifically, participants who reported using their plan more frequently were also more likely to report SI.

Table 1: Logistic Mixed-Effect Model for Suicidal Ideation (SI)

Fixed effects	Estimate(SE)
Intercept	-1.051 (0.001)***
Time	-0.887 (0.001)***
Plan use freq. (within-subject)	0.449 (0.001)***
Plan use freq. (between-subject)	1.021 (0.001)***
Group (CRP)	-0.446 (0.001)***
Plan use freq. (within-subject) $\times$ Group (CRP)	-0.373 (0.001)***
Plan use freq. (between-subject) $\times$ Group (CRP)	-0.917 (0.001)***
Random effects	Variance
Intercept	28.38

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Both within-person plan use frequency and between-person plan use frequency were significantly associated with SI.
- Additionally, individuals with a higher average frequency of plan use overall across the study period were also more likely to report SI.

Table 1: Logistic Mixed-Effect Model for Suicidal Ideation (SI)

Fixed effects	Estimate(SE)
Intercept	-1.051 (0.001)***
Time	-0.887 (0.001)***
Plan use freq. (within-subject)	0.449 (0.001)***
Plan use freq. (between-subject)	1.021 (0.001)***
Group (CRP)	-0.44 (0.001)***
Plan use freq. (within-subject) $\times$ Group (CRP)	-0.373 (0.001)***
Plan use freq. (between-subject) $\times$ Group (CRP)	-0.917 (0.001)***
Random effects	Variance
Intercept	28.38

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- Individuals in the CRP group were less likely to endorse SI compared to the self-guided SP group.

Table 1: Logistic Mixed-Effect Model for Suicidal Ideation (SI)

Fixed effects	Estimate(SE)
Intercept	-1.051 (0.001)***
Time	-0.887 (0.001)***
Plan use freq. (within-subject)	0.449 (0.001)***
Plan use freq. (between-subject)	1.021 (0.001)***
Group (CRP)	-0.446 (0.001)***
Plan use freq. (within-subject) $\times$ Group (CRP)	-0.373 (0.001)***
Plan use freq. (between-subject) $\times$ Group (CRP)	-0.917 (0.001)***
Random effects	Variance
Intercept	28.38

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- There were significant moderation effects.
- These negative interaction coefficients indicate that CRP participants reported lower odds of SI at the same frequency of plan use both on a daily basis (within-person) and on average across participants (between-person).

Conclusion

This study examine whether daily suicidal experiences varied as a function of CRP versus self-guided SP use in the real world. These findings suggest that the two interventions have different effects on suicidal ideation in real life, with CRP use linked to lower risk of suicidal ideation compared to self-guided SP use. Findings have important implications for how providers can deliver safety planning type-interventions to enhance their use by patients in the real world.

Questions?

References

- Aaron T Beck, Robert A Steer, and William F Ranieri. Scale for suicide ideation: Psychometric properties of a self-report version. *Journal of Clinical Psychology*, 44(4):499–505, 1988.
- World Health Organization. *Suicide worldwide in 2019: Global health estimates*. World Health Organization, 2021.